

VERO BEACH, FL, USA

WMO: 747930

Lat: 27.651N

Lon: 80.420W

Elev: 28

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 12843

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	38.8	43.2	25.9	19.9	46.2	31.6	26.0	50.4	24.2	69.9	21.0	65.3	7.0	320	0.389

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.3	92.1	77.2	90.8	77.5	89.9	77.5	80.4	87.9	79.8	87.4	79.1	86.5	10.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.6	148.7	84.1	77.4	142.8	83.6	77.0	140.6	83.3	43.7	87.9	43.1	86.8	42.5	86.9	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.4	18.5	16.9	DB	33.4	96.1	3.4	1.7	30.9	97.3	28.9	98.3	27.0	99.3	24.5	100.6
			WB	30.0	81.9	3.9	1.1	27.2	82.7	25.0	83.3	22.8	84.0	20.0	84.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	73.6	62.3	65.0	68.0	72.3	76.9	80.6	81.8	81.9	80.8	76.7	70.1	65.9
	DBStd	8.72	8.08	7.59	6.75	5.11	3.51	2.36	2.02	1.82	2.13	4.85	6.20	7.84
	HDD50	17	10	3	0	0	0	0	0	0	0	0	0	4
	HDD65	416	147	88	51	10	0	0	0	0	0	4	27	89
	CDD50	8621	392	422	559	670	834	919	986	990	925	826	602	496
	CDD65	3544	64	87	144	229	369	469	521	525	475	365	179	117
	CDH74	31022	288	465	861	1627	3185	4523	5446	5423	4558	3084	1035	527
	CDH80	10157	24	65	163	352	933	1701	2240	2172	1570	752	139	46
Wind	WSAvg	8.1	8.4	8.6	9.3	9.3	8.7	7.4	6.6	6.6	7.2	8.5	8.3	7.9
Precipitation	PrecAvg	52.2	2.8	2.9	3.6	2.6	2.9	6.3	6.0	6.5	7.0	4.8	3.5	2.0
	PrecMax	66.7	8.9	6.9	12.8	10.4	5.7	20.7	11.5	11.6	15.4	12.4	11.8	5.4
	PrecMin	37.8	0.3	0.0	0.2	0.0	0.1	0.3	1.3	3.3	1.7	0.7	0.4	0.2
	PrecStd	8.5	2.1	2.0	3.0	2.6	1.6	4.6	3.0	2.7	3.4	3.4	2.6	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.6	85.2	88.0	89.8	91.9	93.9	94.1	93.3	92.0	90.0	86.5	83.7
		MCWB	71.0	70.7	71.8	71.7	74.0	76.2	77.5	77.7	77.4	76.8	74.8	73.4
	2%	DB	79.7	81.9	84.1	86.3	89.5	91.3	91.4	91.3	90.1	87.6	83.6	81.2
		MCWB	69.5	70.4	70.6	71.9	73.7	77.1	78.0	78.3	77.7	76.6	73.8	72.0
	5%	DB	77.3	79.4	81.4	83.8	86.9	89.9	90.4	90.2	88.4	85.7	81.3	79.3
		MCWB	68.5	69.6	69.7	71.4	73.9	77.2	78.1	78.4	77.5	75.8	72.0	70.7
	10%	DB	74.8	76.8	79.0	81.7	84.6	88.0	89.2	88.9	87.4	84.0	79.4	77.1
		MCWB	67.1	68.1	68.4	70.2	73.5	76.9	78.1	78.3	77.2	74.7	70.5	69.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.4	73.9	75.3	76.5	78.1	80.4	81.3	81.3	80.7	79.8	76.9	75.3
		MCDB	78.7	80.0	81.2	83.4	85.6	88.7	88.9	88.8	87.6	85.3	82.8	80.9
	2%	WB	71.5	72.5	73.3	74.9	76.8	79.3	80.3	80.3	79.8	78.4	75.5	73.8
		MCDB	76.5	78.3	79.7	81.8	84.3	87.5	88.0	88.0	86.7	84.0	80.9	78.7
	5%	WB	70.2	71.1	71.9	73.5	75.9	78.4	79.5	79.7	78.8	77.5	74.3	72.7
		MCDB	75.1	77.0	78.5	80.2	83.2	86.3	87.3	87.4	85.5	83.3	79.4	77.3
	10%	WB	68.8	69.8	70.3	72.4	75.1	77.8	78.7	79.0	78.1	76.6	72.9	71.3
		MCDB	73.7	75.2	76.6	79.1	82.1	85.4	86.3	86.4	84.8	82.1	77.8	75.5
Mean Daily Temperature Range	5% DB	MDBR	19.8	19.7	19.4	18.4	16.8	15.6	15.7	15.3	14.1	14.4	16.8	17.8
		MCDBR	19.3	19.9	20.3	19.8	19.5	18.1	17.0	16.6	15.3	14.5	15.9	16.7
		MCWBR	11.3	10.4	9.7	8.6	7.9	7.1	6.8	6.6	6.2	6.5	8.3	9.3
	5% WB	MCDBR	17.2	17.4	18.2	17.2	16.4	16.0	15.8	15.2	14.1	12.8	14.2	14.9
		MCWBR	11.3	9.5	8.8	8.6	7.6	7.1	7.1	6.8	6.4	6.2	8.1	9.2
Clear-Sky Solar Irradiance	taub		0.347	0.355	0.370	0.396	0.426	0.453	0.469	0.458	0.425	0.393	0.362	0.358
	taud		2.534	2.528	2.474	2.377	2.338	2.338	2.287	2.338	2.423	2.483	2.515	2.525
	Ebn at Noon		285	292	293	286	276	267	262	265	272	275	277	276
	Edh at Noon		29	31	35	39	41	41	43	40	36	32	29	28
All-Sky Solar Radiation	RadAvg		1093	1343	1646	1927	1977	1860	1841	1774	1560	1398	1149	992
	RadStd		77	75	97	69	148	161	117	108	82	90	56	68

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (30 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	-56	+272	+209

Nomenclature: See separate page